

2. Level 5 - Level 10

a. Level 5

Level goals : The password for the next level is stored in a file somewhere under the inhere directory and has all of the following properties:

- human-readable
- 1033 bytes in size
- not executable

Explanation :

- The password in this level is stored in a file, the problem is there are so many folders, checking on each of them manually will waste so much time. Instead check them manually. We can find the exact same file like the clue.

Solution :

1. See all of the folders that we have in the “inhere” folder. We can use “ls -la” to see all of them.

```
bandit5@bandit:~/inhere$ ls -la
total 88
drwxr-x--- 22 root bandit5 4096 Jun 24 14:59 .
drwxr-xr-x  3 root root    4096 Jun 24 14:59 ..
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere00
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere01
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere02
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere03
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere04
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere05
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere06
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere07
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere08
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere09
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere10
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere11
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere12
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere13
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere14
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere15
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere16
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere17
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere18
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere19
```

2. To find the exact same file that has the same criteria like the clue. We need to use command find, use find syntax for this case. find + files criteria that we want to find.
3. In this case, I want to find 1033 byte files and also not executable files. So use “find . -type f -size 1033c | ! -executable”.

```
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable
./maybehere07/.file2
```

4. use cat ./file2 to see the password

Password for level 6 : pXa26xhMWaC2SvDotA4r9EgZkulOeSBW

b. Level 6

Level goals : The password for the next level is stored **somewhere on the server** and has all of the following properties:

- owned by user bandit7
- owned by group bandit6
- 33 bytes in size

Explanation : In this case, there is no file/folder that is seen in the folder challenge. But there is a hidden file that is hidden in the challenge server. Instead of using “ls -la”, we can use “ls -la /” to see all of the hidden folder and files.

Solution :

1. Use “ls -la” to see all of the hidden files in the server.

```
bandit6@bandit:~$ ls -la
.  ..  .bash  logout  .bashrc  .profile
bandit6@bandit:~$ ls -la /
total 112
drwxr-xr-x 28 root root 4096 Aug 25 21:07 .
drwxr-xr-x 28 root root 4096 Aug 25 21:07 ..
drwxr-xr-x  2 root root 4096 Jun 24 14:59 behemoth
lrwxrwxrwx  1 root root    7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x  5 root root 4096 Jun 19 05:23 boot
drwxr-xr-x 15 root root 3280 Sep  9 06:52 dev
drwxr-xr-x  7 root root 4096 Jun 24 15:00 drifter
drwxr-xr-x 124 root root 12288 Aug 17 20:51 etc
drwxr-xr-x  3 root root 4096 Jun 24 15:00 formulaone
drwxr-xr-x 150 root root 4096 Jun 24 15:02 home
drwxr-xr-x  9 root root 4096 Jun 24 15:00 krypton
lrwxrwxrwx  1 root root    7 Apr 20 08:46 lib -> usr/lib
lrwxrwxrwx  1 root root    9 Jun 24 14:50 lib32 -> usr/lib32
lrwxrwxrwx  1 root root    9 Apr 20 08:46 lib64 -> usr/lib64
lrwxrwxrwx  1 root root   10 Jun 24 14:50 libx32 -> usr/libx32
drwx----- 2 root root 16384 Jun 19 05:20 lost+found
drwxr-xr-x  3 root root 4096 Jun 24 15:01 manpage
drwxr-xr-x  2 root root 4096 Jun 24 15:01 maze
drwxr-xr-x  2 root root 4096 Jun 19 05:17 media
drwxr-xr-x  2 root root 4096 Jun 19 05:17 mnt
drwxr-xr-x  2 root root 4096 Jun 24 15:02 narnia
drwxr-xr-x  6 root root 4096 Jun 24 14:53 opt
dr-xr-xr-x 707 root root    0 Sep  9 06:50 proc
drwx----- 9 root root 4096 Aug 17 20:53 root
drwxr-xr-x 36 root root 1080 Aug 30 20:25 run
lrwxrwxrwx  1 root root    8 Apr 20 08:46/sbin -> usr/sbin
drwx----- 6 root root 4096 Jun 19 05:24 snap
drwxr-xr-x  2 root root 4096 Jun 19 05:17 srv
dr-xr-xr-x 13 root root    0 Aug 25 21:27 sys
drwxrwx-wt 109 root root 3260 Sep  9 06:52 tmp
drwxr-xr-x 14 root root 4096 Jun 24 14:59 usr
drwxr-xr-x  2 root root 4096 Jun 24 15:02 utumno
drwxr-xr-x 14 root root 4096 Jun 25 12:37 var
drwxr-xr-x  2 root root 4096 Jun 24 15:03 vortex
```

2. In the clue, we know that we need to find a specific file name password, but there is no password seen in the directory. So we need to use command find. We can write

```
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c
find: '/snap': Permission denied
find: '/lost+found': Permission denied
find: '/manpage/manpage3-pw': Permission denied
find: '/drifter/drifter14_src/axTLS': Permission denied
find: '/run/pam_pidns': Permission denied
find: '/run/udisks2': Permission denied
find: '/run/chrony': Permission denied
find: '/run/user/11008': Permission denied
find: '/run/user/11024': Permission denied
find: '/run/user/12002': Permission denied
find: '/run/user/8004': Permission denied
find: '/run/user/20003': Permission denied
find: '/run/user/16000': Permission denied
find: '/run/user/11030': Permission denied
find: '/run/user/11015': Permission denied
find: '/run/user/11011': Permission denied
find: '/run/user/11010': Permission denied
find: '/run/user/11025': Permission denied
find: '/run/user/8005': Permission denied
find: '/run/user/11009': Permission denied
find: '/run/user/11023': Permission denied
find: '/run/user/11004': Permission denied
find: '/run/user/11002': Permission denied
find: '/run/user/11003': Permission denied
find: '/run/user/11014': Permission denied
find: '/run/user/11019': Permission denied
find: '/run/user/11020': Permission denied
find: '/run/user/11001': Permission denied
find: '/run/user/11013': Permission denied
find: '/run/user/11016': Permission denied
find: '/run/user/11006/systemd/inaccessible/dir': Permission denied
find: '/run/user/11012': Permission denied
find: '/run/user/13000': Permission denied
find: '/run/user/11005': Permission denied
find: '/run/user/11000': Permission denied
find: '/run/user/13003': Permission denied
```

```
find: '/var/lib/ubuntu-advantage/apt-esm/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/amazon': Permission denied
find: '/var/lib/udisks2': Permission denied
find: '/var/lib/dpkg/info/bandit7.password'
find: '/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/private': Permission denied
find: '/var/cache/apparmor/147ee081.0': Permission denied
find: '/var/cache/apparmor/acl4050.0': Permission denied
find: '/var/cache/apt/archives/partial': Permission denied
find: '/var/cache/pollinate': Permission denied
find: '/var/cache/ldconfig': Permission denied
find: '/var/cache/private': Permission denied
find: '/var/spool/rsyslog': Permission denied
find: '/var/spool/cron/crontabs': Permission denied
find: '/var/spool/bandit24': Permission denied
find: '/var/crash': Permission denied
find: '/var/log': Permission denied
find: '/var/tmp': Permission denied
find: '/etc/xinetd.d': Permission denied
find: '/etc/credstore.encrypted': Permission denied
find: '/etc/credstore': Permission denied
find: '/etc/multipath': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/stunnel': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/ssl/private': Permission denied
find: '/boot/efi': Permission denied
find: '/boot/lost+found': Permission denied
find: '/proc/tty/driver': Permission denied
find: '/proc/1/task/1/fd': Permission denied
find: '/proc/1/task/1/fdinfo': Permission denied
find: '/proc/1/task/1/ns': Permission denied
find: '/proc/1/fd': Permission denied
find: '/proc/1/map_files': Permission denied
find: '/proc/1/fdinfo': Permission denied
```

3. If we see manually, we can see a strange directory path. /var/lib/dpkg/info/bandit7.password . So all we need to do is cat /var/lib/dpkg/info/bandit7.password

```
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3
bandit6@bandit:~$
```

Password for level 7 : Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3

c. Level 7

Level goal : The password for the next level is stored in the file data.txt next to the word millionth

Explanation : In this case, we know the clue is the password is in the file name “data.txt”. The problem is there are so many words, if we check them manually. It will take a lot of time, we already know there is a file named “data.txt”. All we need to do is combine cat and grep “millionth” string.

Solution :

1. We use this command `cat data.txt | grep "millionth"`, to find a specific string named "millionth".

```
bandit7@bandit:~$ cat data.txt | grep "millionth"
millionth      VR1ljMayciFxbnUokuQmJFw6QC9VKtub
bandit7@bandit:~$
```

Password for level 8 : VR1ljMayciFxbnUokuQmJFw6QC9VKtub

d. Level 8

Level goal : The password for the next level is stored in the file data.txt and is the only line of text that occurs only once

Explanation : Same like the previous case, there is so much non-readable string. But there is a clue, the password only occurs once.

Solution :

1. The first thing we want to do is see what is inside of data.txt. Write cat data.txt

[illegible]

2. There is so much string in this data, so we need to sort all of them and find the unique one. Use “Sort * | uniq -u” to find the only unique string. If you find it, then that’s the password.

```
bandit8@bandit:~$ sort * | uniq -u
Ejm0SvuAu7sGAHqHVcBDPirRe9T03kxl
bandit8@bandit:~$
```

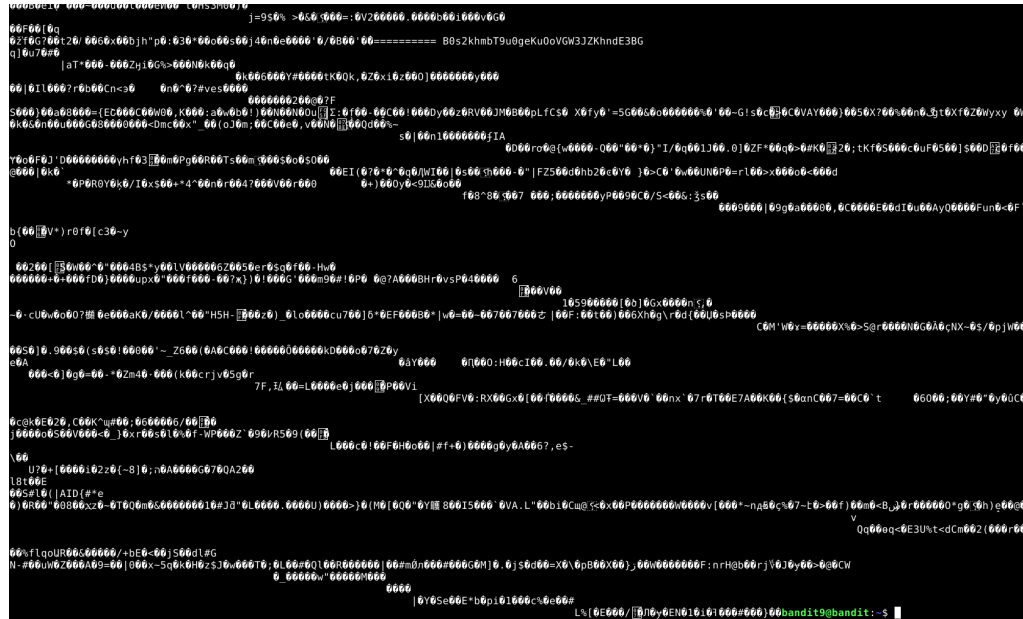
Password for level 9 : EimOSvuAu7sGAHqHVcBDPirRe9T03kxI

e. Level 9

Level goal : The password for the next level is stored in the file data.txt in one of the few human-readable strings, preceded by several '=' characters.

Explanation :

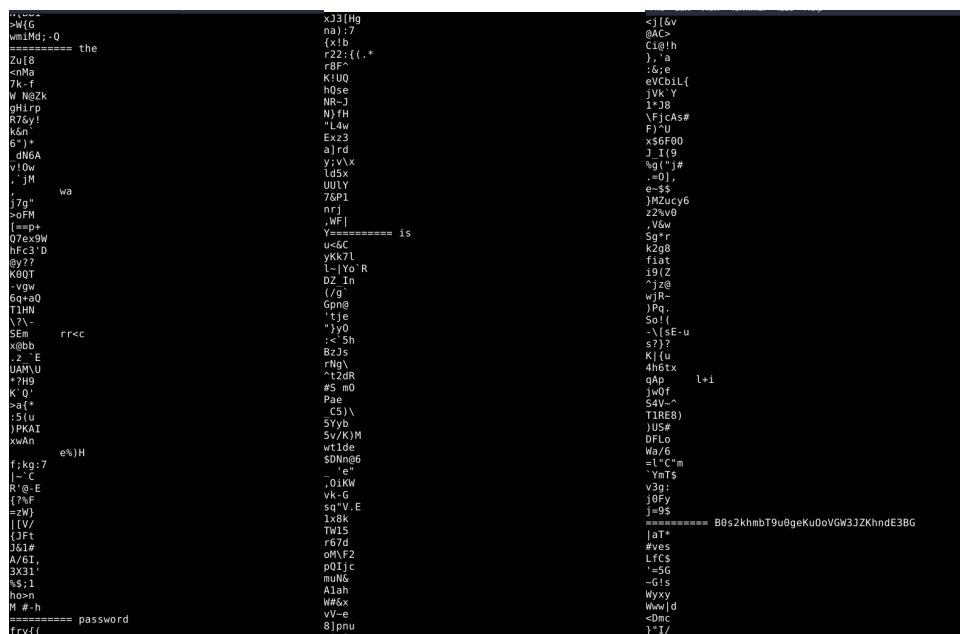
In this case, if we try to see the inside of the file “data.txt”. We only see unreadable strings.



But in the clue, we can see that the password is human-readable strings and also between “=”.

Solution :

1. Use string filename, to convert file type into strings. So we can see the string style of this file, using "string data.txt".



There is a unique pattern, that if we use string data.txt we can see a clue that says "B0s2khmbT9u0geKuOoVGW3JZKhndE3BG" is the password.

Password for level 10 : B0s2khmbT9u0geKuOoVGW3JZKhndE3BG

f. Level 10

Level goal : The password for the next level is stored in the file data.txt, which contains base64 encoded data

Explanation : inside of data.txt, we can see an encoded data of the password. Then how we can get the password? we need to decode the data again and we will get the password. The clue say they use base64 to encoded data.txt, so to able see the password, we need to decode with base64 again.

Solution :

1. Use base64 to decoded, "base64 -d data.txt".

```
bandit10@bandit:~$ base64 -d data.txt
The password is pYf0Y6HwUsDj5rL9UvyhU7MCmv8vN5Ro
```

Password for level 11 : pYf0Y6HwUsDj5rL9UvyhU7MCmv8vN5Ro